MILITARY
SPECIFICATION

MIL-A-11892(SigC)
31 March 1952

ANTENNA GROUP OA-249( )/GRC-10
(Components of)

1. SCOPE

1.1 This specification covers the requirements for the following major components of Antenna Group OA-249( )/GRC-10.

Case, Antenna CY-1252( )/GRC-10
Antenna AS-620( )/GRC-10
Mast AB-301( )/U

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standard, drawing and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

GGG-H-86	Hammers, Mauls and Sledges
QQ-M-151	Metals; General Specification for inspection of
V-T-276	Thread, Cotton
QQ-A-325	Aluminum Alloy 618 Bars, Rods, Shapes and Wire, Rolled or Drawn
QQ-A-327	Aluminum Alloy 618, Plate and Sheet
QQ-S-571	Solder, Soft (Tin, Tin-Lead and Lead-Silver)
QQ-A-601	Aluminum-Base-Alloys: Sand Castings
T-R-601	Rope, Manila
TT-C-595	Colors; for Ready-Mixed Paints
MM-L-736	Lumber and Timber, Hardwood
WW-T-789	Tubing, Aluminum Alloy

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JAN-C-17	Cables, Coaxial and Twin Conductor, For Radio Frequency
MIL-P-66	Plywood; Flat-panel
JAN-P-105	Packaging and Packing for Overseas Shipment-Boxes; Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment-Boxes; Wood, Nailed
JAN-P-108	Packaging and Packing for Overseas Shipment-Boxes; Fiberboard (V-Board and W-Board), Exterior and Interior

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MILITARY (Continued)

MIL-P-116	Packaging and Packing for Overseas Shipment - Preservation, Methods of
JAN-P-117	Packaging and Packing for Overseas Shipment - Bags, Interior Packaging
JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water Resistant
JAN-P-397	Adhesive, Room-Temperature and Intermediate Setting Resin (Phenol, Resorcinol, and Melamine Base) (For Wood)
JAN-W-530	Webbing, Cotton, Natural or in Colors
JAN-A-669	Anti-Seize Compound, White Lead Base, General Purpose (for Thread Fittings)
MIL-R-3065	Rubber and Synthetic Rubber Compounds: General Purpose (Except Tubes, Inner Tubes, Sponge Rubber and Hard Rubber)
MIL-P-3115	Plastic-Material Laminated Thermosetting Sheets, Paper-Base, Phenolic-Resin
MIL-A-5433	Adhesive: Application of Room-Temperature and Intermediate-Temperature Setting Resin (Phenol, Resorcinol and Melamine Base)
MIL-D-10861	Duck; Cotton Plied-Yarns (Army, Numbered and Special Use
MIL-C-11436	Chromium Gray Plated
MIL-T-16070	Treatment, Mildew Resistant for Rope (Manila)

U. S. ARMY

19-10	Filler, Interior Packing, Fiberboard, Corrugated, Single and Double Faced for General Packing Purposes
71-4944	Marking (For Signal Corps Equipment)
72-53	Finishes (For Signal Corps Equipment)

STANDARDS

MILITARY

MIL-STD-105	Sampling Inspection Tables for Inspection by Attributes
MIL-STD-129	Marking of Shipments

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DRAWINGS

SIGNAL CORPS

SC-D-1155	Mast Section MS-44
SC-A-28049	Guy Plate MX-378/U Detail
SC-D-28051	Stake GP-101/U Assembly and Detail

(Copies of specifications, standard, and drawings required by contractor in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

NATIONAL BUREAU OF STANDARDS

Handbook H-28	Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

AMERICAN STANDARDS ASSOCIATION PUBLICATIONS:

B.18.6	Slotted and Recessed Head Screws

(Application for copies should be addressed to the American Standard Association, Incorporated, 70 East Forty-Fifth Street, New York 17, New York.).

AMERICAN SOCIETY FOR TESTING MATERIALS

A-123	Zinc (Hot Galvanized) Coating in Structural Steel Shapes, Plates and Bars, and Their Products

(Application for copies should be addressed to The Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

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3. REQUIREMENTS

3.1 Model.- The referenced model shall be understood as the one which, in accordance with the invitation for bids, will be available for inspection by prospective bidders and will be lent to the contractor. In case of conflict between any feature of the model and the requirements of this specification, the requirements of this specification shall govern. Unless otherwise specified herein, the equipment shall conform to the following:

3.1.1 Construction.- Construction of the equipment shall conform to the model.

3.1.2 Functions.- The equipment shall incorporate all functions provided by the model.

3.1.3 Performance.- Performance of the equipment shall be at least equal to that of the model. When any performance characteristic of the model is not specified herein, the following shall apply: As soon as the model is received by the contractor, he shall adjust it for optimum performance and prepare to conduct all measurements necessary to determine the performance characteristics which are not specified herein. The contracting officer shall be notified of the date when the measurements are to be taken, sufficiently in advance, to send a representative to witness them. Measurements shall be taken by the contractor and three record copies thereof furnished to the contracting officer.

3.2 Preproduction samples.-

3.2.1 Quantities of preproduction samples.- After award of contract, preproduction samples shall be furnished as shown below. (See 6.3.2.)

Four complete equipments

3.2.2 Compliance with specified requirements.-

3.2.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.2.2.3 for exception.)

3.2.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirements. After being released to the contractor (See 3.2.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.2.3 When deviation from 3.2.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement(see 3.2.2.1) describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.2.3 Test data.- The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like.)

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3.2.4 Reference standards.- After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of specified requirements.

3.3 Parts, materials and processes.- All parts, materials and processes not specifically mentioned herein shall be in accordance with those specifications listed in Section 2 covering such parts, materials and processes. The selection of class, grade, or type shall be such that, when mounted in the equipment, the part or material will perform its intended function.

3.3.1 Metals.- The metals used shall be of the proper alloy and hardness necessary to provide the required strength and rigidity. Materials shall be of corrosion-resisting types or shall be treated in accordance with Specification No. 72-53. Dis-similar metals used in contact with each other shall be carefully selected, in accordance with the requirements of Specification No. 72-53, to prevent electrolytic corrosion in the presence of moisture.

3.3.1.1 Sand castings.- All aluminum sand castings shall be Class 4 in accordance with Specification QQ-A-601.

3.3.2 Solder, soft.- Soft solder shall be in accordance with Specification QQ-S-571 and, where used for electrical connections, shall have a minimum tin content of 60.0 percent by weight unless a lower percentage is approved, by the contracting officer, for the specific application.

3.3.2.1 Flux.- Flux, including that used in flux-cored solder, shall be rosin or rosin and alcohol. The rosin shall be grade "WW" (water white), or better. Where soldered materials are such that the use of rosin and alcohol is impracticable, the contractor shall secure permission of the contracting officer before using alternate fluxes and cleaning agents. Any flux other than rosin shall be chemically neutralized and removed.

3.3.2.2 Soldering.- Soldering shall be done neatly. Wherever practicable, the excess flux shall be removed with alcohol or other suitable solvent after soldering. Solder shall not have crystallized, between inadequately heated, or overheated. Sharp points or rough surfaces resulting from overheating or insufficient heating will not be acceptable. The solder shall feather out to a thin edge, indicating proper flowing and wetting action. Where insulating material is subjected to heating during the soldering operation, it shall be undamaged, and parts fastened thereto shall not become loosened. Where soldered connections are made, the bared wire shall be coated with solder to prevent corrosion.

3.3.3 Plastic materials.- Where not machined, plastic materials shall have the original smooth or polished surfaces unless objectionable glare makes a dull surface more desirable. All surfaces that have been sawed, cut or otherwise machined shall be reasonably smooth. Lined or cotton based plastic shall not be used for electrical applications.

3.3.4 Riveting.- After completion of the riveting operations the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their adjacent surfaces.

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3.3.5 Welding.- Surfaces to be welded shall be cleaned in accordance with good commercial practice to be free from rust, scale, paint, grease, and other foregin material. Where embrittlement of the parent metal is likely to result, preheating or annealing shall be employed. Welds shall be of ample size with thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock-marks, bruises, or other harmful defects. Where undesirable internal stresses may result from welding, the welded assembly shall be stress relieved. All complex welded assemblies of high strength aluminum or magnesium must be stress relieved. All complex welded assemblies of high strength aluminum or magnesium must be stress relieved. Welds shall be cleaned of scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed. Where tack or spot welds are employed to secure parts permanently, the number of welds shall be sufficient to provide adequate strength and at least 2 welds shall be used to hold each part.

(a) Inert gas shielded arc welding is the preferred welding method for aluminum and magnesium.

(b) Wherever possible the welded joint should be so designed so that the weld does not have to be ground or flushed off.

(c) Unless the design dictates otherwise, spot welds in aluminum or magnesium should be confined to lightly stressed non-critical structures.

3.3.6 Mounting or assembly of parts made of soft material.- Wherever practicable parts made of aluminum, magnesium, plastic, or other soft material shall be mounted or assembled by one or more of the following means:

3.3.6.1 Tapped plated steel bushing, nut or insert that is permanently secured to the structure, where frequent disassembly, and reassembly will be necessary.

3.3.6.2 Through-bolt secured by a stopnut. A flatwasher shall be used under the head of the bolt unless the shape of the head is such that the flatwasher is unnecessary.

3.3.6.3 Tapped hole that is protected against wear from frequent assembly and disassembly by "Heli-Coil" thread inserts as made by the Heli-Coil Corp., Long Island City, New York, or equal.

3.3.6.4 Stud in a tapped hole. The tapped hole shall be of a coarser pitch than the free end of the stud and shall be such that maximum possible strength is provided to hold the stud. The "Lok-Thud" as made by the National Screw and Mfg. Co., Cleveland, Ohio, or equal, is a preferred stud thread.

3.3.6.5 All inserts that are mounted in aluminum castings shall be stainless steel.

3.3.7 Mounting or assembly of parts made of brittle material.- The brittle assembly shall not be over-constrained by the securing means. Not more than three point mounting shall not be used unless the surfaces are accurately machined for mating.

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3.3.8 Screw threads, threaded parts, and related items.-

3.3.8.1 Screw threads.- Screw threads shall conform to Handbook H28 and shall be class 2 fit. Except forholding labels or similar lightweight parts not used for structural purposes or critical applications, threads in aluminum or other soft material shall not be employed where other means of fastening are practicable and, in particular, shall not be used when frequent disassembly and reassembly may be necessary. When considered necessary by the Government inspector, based on his inspection of the equipment or observation of the contractor's production methods, the contractor shall use a torque measuring or limiting device for tightening threaded parts. When such devices are used, the values of torque shown in Table I shall be applied to steel screws. When torque measuring or limiting devices are used for other sizes, pitch, or materials, the torque shall be the design value established by the contractor for the particular application.

Table I - Torques

Thread	Torque (in. -lb.)
6-32	9 to 12
8-32	18 to 24
10-24	29 to 34
10-32	32 to 40
12-24	42 to 56

3.3.8.2 Thread engagement and extension.- Length of thread engagement in tapped holes shall be at least equal to that provided by the corresponding semi-finished regular nut, when the hole is tapped in steel or hard brass. The engagement shall be at least 50 percent greater if the hole must be tapped in aluminum, magnesium, or other soft materials. Where permitted by design of the equipment, screws and bolts shall extend at least 1-1/2 threads beyond the nut.

3.3.8.3 Material and finish for fastening devices.- Fastening devices (screws, bolts, nuts, washers, etc) shall be of steel unless electrical considerations require use of other materials, and shall be finished in accordance with Specification 72-53.

3.3.8.4 Screw sizes.- Where practicable, screws shall be no smaller than No. 6 (0.138-inch maximum body diameter). Odd-numbered screws shall not be used.

3.3.8.5 Shapes of screws and bolts.- Screws and bolts shall conform in shape to Standard B18.6. Where practicable, No. 6 screws (0.138-inch maximum body diameter) or larger shall have slotted-hexagon or fillister heads. Hexagon heads shall be "clipped". Screw driver slots shall be milled.

3.3.8.6 Self-tapping screws.- Thread-cutting screws may be used where construction is improved thereby and they will not require loosening or removal during operation or maintenance of the equipment. Chips formed by self-tapping screws in through-holes shall be removed. Self-tapping sheet-metal screws shall not be used.

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3.3.8.7 Set screws.- Set screws shall be used only where uniquely required by design of the equipment. Set screws shall be of the hardened, hexagon-socket type and preferably shall be No. 6 (0.138-inch maximum body diameter) or No. 10 (0.190-inch maximum body diameter). Cone-pointed set screws may be used where the engaging surfaces are suitably countersunk to receive the point, otherwise sup pointed set screws shall be used. Flat surfaces shall be provided for engagement of cup-pointed screws when the part is not adjustable in angular relationship to the shaft on which it is secured. Where two set screws are used, the angle between the major axes shall be not less than 90 degrees and not more than 120 degrees.

3.3.8.8 Anchor nuts and nut plates.- Where practicable, anchor nuts or nut plates shall be used for threaded engagement in light sheet metal.

3.3.8.9 Anti-seize compound.- Anti-seize compound conforming to Specification JAN-A-669 shall be used to prevent seizing of threads where the threaded surface is of aluminum or magnesium.

3.3.8.10 Embrittlement.- High-carbon fasteners and associated spring devices shall be baked to relieve hydrogen embrittlement, immediately after processing, if the plating, cleaning, or fluxing operations produce such effect.

3.3.9 Finishes.- Where not otherwise specified, all surfaces shall be suitable protected from deterioration by finishes of either Type I or Type II in accordance with Specification 72-53, as determined by the exposure to which the surfaces will be subjected. All surfaces used for electrical contact shall be plated in accordance with the applicable requirements of Specification 72-53.

3.3.9.1 Color.- The color of the painted surface of each component of the equipment, with the exception of cables and dipole elements, shall be semi-gloss olive drab, Shade No. 2430, in accordance with Specification TT-C-595.

3.3.10 Cleaning.- The equipment shall be thoroughly cleaned and shall be free of foreign material. Welds shall be thoroughly cleaned of flux and oxidation products. All traces of corrosive compounds shall be removed to prevent subsequent corrosion of the parts, or of other parts adjacent thereto.

3.4 Ruggedness and reliability.- All parts and details of construction shall have the maximum of ruggedness and reliability under all operating conditions. The equipment shall be so constructed that it will withstand long periods of service and repeated rough handling in transportation, with a minimum of damage and/or readjustment.

3.5 Interchangeability.- All components, subassemblies and parts shall be mechanically and electrically interchangeable with all corresponding components, sub-assemblies and parts of the equipment on order.

3.6 Marking.-

3.6.1 Nomenclature.- Complete equipments and parts to which Army-Navy or Signal Corps Nomenclature has been assigned shall bear identification marking in accordance with Specification No. 71-4944 (see D-1a(2) of Specification 71-4944).

3.6.2 Marking of standard parts.- Parts covered by standard specifications shall be marked as required by those specifications.

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3.6.3 Other parts not covered by standard specifications, including mechanical parts, shall be makred, wherever space permits, with the manufacturer's name (or trademark), catalog or other identification number, value and rating (if practicable) and other pertinent information. If space is limited, as much as practicable of the required information shall be included, preference being in the other given above. Sub-assemblies that are replaceable as a unit shall be marked in the same manner but the marking shall be such is not to be confused with marking of any replaceable part thereof.

3.7 Tests.- Antenna Group OA-249( )/GRC-10 shall be capable of meeting all the tests of Section 4.

3.8 General characteristics.- The equipment covered by this specification shall be capable of operation over the frequency range of from 54 to 71 mc and shall be horizontally polarized. It shall consist of the components as specified in the bid request and contract.

3.9 Case, Antenna CY-1252( )/GRC-10.- Case, Antenna CY-1252( )/GRC-10 shall be constructed in accordance with the model. The case shall incorporate a carrying rack so that two complete arrays can be transported as a unit without the outer transit case. The plywood used in the construction of the case shall be Douglas Fir, Exterior not lower than grade A-C in accordance with Specification MIL-P-66.

3.10 Antenna AS-620( )/GRC-10.-

3.10.1 Construction.- Antenna AS-620( )/GRC-10 shall be constructed in accordance with the model.

3.10.2 Antenna elements.- Antenna elements and the boom shall be aluminum alloy, Condition T, in accordance with Specification WW-T-789.

3.10.2.1 Plating of antenna elements.- The movable ends of the antenna elements shall be plated gray chrome in accordance with Specification MIL-C-11436(SigC).

3.10.2.2 Hardware.- Wing bolts and inserts shall be made of stainless steel. (Stainless steel shall be the non-nickel bearing type)

3.10.2.3 Color coding.- The adjustable parts of the antenna array shall be marked and color coded in accordance with the model of so as to facilitate reference to the frequency setting chart of 3.10.2.4.

3.10.2.4 Frequency setting charts.- Frequency setting charts shall be provided to indicate setting of element lengths and boom spacing for maximum gain at each megacycle point over the frequency range of the equipment.

3.10.3 Rubber gasketing.- Rubber gaskets shall be 1/16" thick and shall be Grace SC-614 ABFF in accordance with Specification MIL-R-3065 with the following exceptions:

3.10.3.1 The maximum elongation shall be 250 percent.

3.10.3.2 Compression, Suffix B, shall be not greater than 20 Percent at the test condition of 70 hours at 212 F.

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3.10.3.3 The oil resistance test is not applicable.

3.10.4 Mounting.- The antenna shall be capable of being mounted on Mast AB-301( )/U

3.10.5 Cables.-

3.10.5.1 One cable assembly shall be provided with each Antenna AS-620( )/GRC-10 consisting of 65 feet of Radio Frequency Cable RC-8A/U in accordance with Specification JAN-C-17 with Plug Connector UG-573/U, Industrial Products Co., Part/Dwg IPC-8000, or equal, on each end.

3.10.5.2 One cable assembly shall be provided with each Antenna AS-620( )/GRC-10 consisting of 35 feet of Radio Frequency Cable RG-8A/U in accordance with Specification JAN-C-17 with Plug Connector UG-573/U Industrial Products Co., Part/Dwg IPC-8000, or equal on each end.

3.10.6 Adapters.- Adapter Connector UG-643/U Industrial Products Co., Part/Dwg IPC-83125 or equal in the quantities called for in the bid request and contract shall be provided to connect radio frequency cables for increased lengths.

3.10.7 Cord and guy identification.- All cords and cables shall be identified by a band into which has been stamped the identifying Signal Corps or JAN type number. The bands shall be placed on each end of the cable. Guys shall be identified by means of a metal tag attached to the snap end.

3.11 Mast AB-301( )/U.-

3.11.1 Mast sections.- Mast sections shall be constructed in accordance with Signal Corps Draving SC-D-1155 with the following exceptions:

3.11.1.1 A 1/4 inch aluminum pin shall be riveted 5-7/16 inches from the large end of the mast section. A 0.260 inch to 0.270 inch double slot shall be provided in line with the pin on the small end of the mast section for keying the section, allowing 1/16 inch over travel on the pin when the sections are assembled. The rivet pin shall be aluminum in accordance with Specification QQ-A-325.

3.11.2 Guy assemblies.- Guy assemblies in the quantities and sizes called for in the bid request and contract shall be constructed in accordance with their respective models. Each assembled guy shall have a minimum breaking strength of 300 pounds.

3.11.2.1 Nylon.- The nylon used shall be stabilized filament nylon, 3/16" standard lay with an average breaking strength of 800 pounds, as supplied by Columbia Rope Co., Auburn, N. Y. or equal.

3.11.2.2 Hook.- The hook shall be 5/8" eye, North and Judd #5308 or equal.

3.11.3 Mast base.- The mast base shall be constructed in accordance with the model. The stake portion shall be GP-101/U in accordance with Signal Corps Draving SC-D-28051. The swivel pin shall be stainless steel. The cotter pin shall be of fabricated of non-corrosive material i.e. brass, phosphor bronze or stainless steel. (Stainless steel shall be the non-nickel bearing type)

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3.11.4 Guy plate.- The guy plate shall be according to model. Rollers, pins and stud, shall be of stainless steel. (Stainless steel shall be the non-nickel bearing type)

3.11.5 Gin pole cap.- The Gin pole cap shall be as per Signal Corps Drawing SC-A-28049 with hook attached as per model. Hook shall be type number 5308, 5/8" eye as manufactured by North and Judd, or equal.

3.11.6 Rope assembly.- The rope assembly shall be in accordance with model. Rope shall be Manilla, per Federal Specification T-R-601 and shall be mildew and moisture-resistant, per Specification MIL-T-16070.

3.11.7 Stake.- The stake shall be in accordance with model and shall be hot zinc galvanized in accordance with ASTM Standard A-123-33.

3.11.8 Case, Cable Assembly CY-1251( )/U.- Case, Cable Assembly CY-1251( )/U shall be constructed in accordance with the model.

3.11.9 Mast base plates.- Mast base plates shall be in accordance with the model.

3.12 Electrical characteristics.- The equipment shall have the following electrical characteristics:

3.12.1 Gain.- In the horizontal plane, horizontally polarized, the ratio of the forward gain of the antenna, over the gain of a half wave dipole for the frequency range 54 to 71 mc shall be at least 6 db, voltage ratio.

3.12.2 Pattern.- The antenna pattern of the horizontally polarized array for the frequency range 54 to 71 mc shall not exceed 90 degrees to the half power points.

3.12.3 Lobe ratio.- The ratio of the maximum field strength of the front lobe to the maximum field strength of any minor lobe, throughout the frequency range of 54 to 71 mc shall be 12 db or better, voltage ratio.

3.12.4 Voltage standing wave ratio.- The voltage standing wave ratio for the frequency range 54 to 71 mc shall be 2.0:1 or better when measured at the end of 65 foot length of RG-8A/U cable.

3.12.5 Frequency coverage.- The equipment shall cover the frequency spectrum of from 54 to 71 mc with settings at each megacycle point within its frequency range.

3.13 Workmanship.- Components, subassemblies, and parts shall be manufactured in a thoroughly workmanlike manner.

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4. SAMPLING, INSPECTION AND TEST PROCEDURE

4.1 Inspection, general.- Inspection shall consist of Group A and C inspection as listed in Table II, and as specified in 4.1.1 through 4.1.3 (See 6.3.1). When practical, Group C inspection shall be performed on sample units that have been subjected to and have met Group A inspection. Samples for inspection shall be selected at random by the Government inspector.

4.1.1 Group A inspection.- At the discretion of the Government inspector, Group A inspection shall be conducted by either or both of the following methods:

(a) 100 per cent inspection.

(b) Statistical sampling in accordance with MIL-STD-105. AQL's (Acceptable Quality Levels) shall be as specified by the contracting officer.

4.1.2 Group C inspection.- Group C inspection shall be performed as follows:

4.1.2.1 Selection of samples.- Samples required shall be selected at random by the Government inspector in accordance with the following:

Lot size	Inspection	
	n	c
9-65	3	1
66-300	3	1
301-800	3	1
801-1300	7	2
1301-3200	7	2
3201-8000	11	3

n = Sample Size
c = Acceptance Number

4.1.2.2 Failure.- If a sample fails on Group C inspection, the contractor shall immediately investiage the cause of failure and take corrective action to the satisfaction of the Government inspector.

Table II - Inspection Procedures

Inspection	Test Paragraph	Preproduction	Group A	Group C
Visual and Mechanical	4.2	X	X	-
Continuity test	4.3	X	X	-
Performance test	4.4	X	-	X
Bounce test	4.5.1	X	-	-
Shock test	4.5.2	X	-	-
Salt Spray test	4.5.3	X	-	-

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4.2 Visual and mechanical.- A thorough inspection of each equipment on the order shall be made for mechanical fit, loose nuts and screws, poor workmanship and other defects. Controls and fastening devices shall be checked for proper mechanical operation. Wiring, soldered connections, ground connection, finishes and the like shall be checked for satisfactory workmanship.

4.3 Continuity test.- The equipment shall be tested for continuity to ascertain that all circuits are wired properly and that plugs and jack connections make good electrical contact, and there are no unwanted short or open circuits.

4.4 Performance test.- Tests shall be performed to determine that the equipment meets the requirements of Section 3. These performance tests will be conducted at the contractor's plant or at a Government laboratory at the discretion of the contracting officer.

4.4.1 Antenna assembly test location.- The transmitting antenna assembly which is the antenna assembly under test shall be mounted on a 40 foot mast. The receiving antenna assembly under test shall be mounted on a 40 foot mast. The receiving assembly, polarized the same as the transmitting antenna assembly, and consisting of a half wave dipole, shall be mounted on a 40 foot mast and used to establish a reference field strength. The antenna assembly under test and the receiving antenna assembly shall be placed at a distance of at least 200 feet apart in level locations that are clear of obstructions in all directions for at least 100 yards. Provision shall be made to eliminate the effect of multipath arrival of the test signal at the receiving end.

4.4.2 Test frequencies.- The frequencies at which the equipment shall be tested for conformance with 3.12 are as follows: 54, 58, 62, 66, 70 megacycles.

4.5 Preproduction tests.- Tests shall be performed to determine the equipment meets the requirements of Section 3.

4.5.1 Bounce test.- The equipment in its transit chest shall be placed on the table of the package tester as made by L.A.B. Corporation, Summit, N. J., or equal. The equipment may be loosely constrained by wooden cleats. The package tester, shafts in phase, shall be run at a speed of 285 rpm for three hours, the top, bottom, and one side of the case shall rest on the table, each side for one hour duration.

4.5.2 Shock test.- The equipment in its transit chest shall be dropped a distance of 3 feet on each corner, edge and face. Drops may be made from a quick release hook, by an Acme Drop Tester, by a 10° inclined plane (Conbur) tester or by a basket type pendulum tester. The barrier or floor receiving the impact shall be 2-inch fir backed by concrete or by a rigid steel frame.

4.5.3 Salt spray test.- The folded dipole feed assembly shall be subjected to 72-hour salt spray test, conditions specified under Specification QQ-M-151A. The feed assembly shall be removed from the chamber and tested, rinsed with tap water and tested again. The test shall consist of a leakage check across the input cable connection using a Megger leakage tester (150 volts dc applied). The resistance shall be no less than 100,000 ohms.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment.- Antenna Group OA-249( )/GRC-10 shall be preserved and packaged in accordance with best commercial practice. The equipment shall be adequately cushioned, blocked, and braced using suitable materials and containers as required, in such a manner as to afford maximum protection from hazards normally encountered in storage and transit.

5.1.2 For overseas shipment.- Preserve and package Antenna Group OA-249( )/GRC-10 in accordance with procedures specified for the designated methods as prescribed in Specification JAN-P-116, and further described in the following paragraphs.

5.1.2.1 Arrays in case.- Package each array in case for mechanical and physical protection by securing the arrays in the case, and spare array sections in the lid of the case. Close case and secure fastenings. Cushion case on all surfaces with cells and/or pads, fabricated of corrugated fiberboard, conforming to Specification 19-10. Place each cushioned case within a close fitting suitable style fiberboard box, conforming to Specification JAN-P-108 for Grade 5. Seal entire box closure with water resistant pressure sensitive tape conforming to Specification JAN-P-127.

5.1.2.2 Array accessories in Bag BG-102.-

5.1.2.2.1 Cables, RF.- Package each cable for mechanical and physical protection by winding each cable into a coil of proportionate dimensions. Tie at 3 places with cotton tape securing the connectors to the coils with the ties.

5.1.2.2.2 Tape.- Package each roll of tape method 1A-9, as follows: Inclose each roll of tape within a close-fitting Type I bag conforming to Specification JAN-P-117 for Class B. Properly seal closure.

5.1.2.2.3 Adapters.- No packaging required.

5.1.2.2.4 Bag BG-102.- Package each Bag BG-102 for mechanical and physical protection as follows: Stow the items, packaged as specified in 5.1.2.2.1, 5.1.2.2.2 and 5.1.2.2.3, within the bags provided. Close bag and secure fastenings. Place the bag within a close fitting suitable style corrugated fiberboard box, conforming to Specification JAN-P-108 for Grade 5. Seal entire box closure with water resistant pressure-sensitive tape.

5.1.2.3 Bags containing mast sections and stakes.- Package each bag containing mast sections and stakes for mechanical and physical protection as follows: Stow and secure the required number of mast sections and stakes within the repropriate sections of the bag. Close covers and secure all bag fastenings and straps. Continue packaging as specified in 5.1.2.2.4.1

5.1.2.4 Mast accessories in bags.-

5.1.2.4.1 Guys.- Package each guy for mechanical and physical protection by winding each guy into a coil of proportionate dimensions. Tie at 3 places with cotton tape. Secure the connectors to the coils with the ties.

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5.1.2.4.2 Hammer, guy plates, mast base plate, mast bases.- No packaging required.

5.1.2.4.3 Bag BG-102.- Package each bag for mechanical and physical portection as follows: Stow the items, packaged as specified in 5.1.2.4.1 and 5.1.2.4.2, within the bag. Continue packaging as specified in 5.1.2.2.4.

5.2 Packing.-

5.2.1 For domestic shipment.- Antenna Group OA-249( )/GRC-10, packaged as specified in 5.1.1, shall be packed in substantial containers of the type, size, and kind commonly used for the purpose, in such a manner as to insure acceptance at and safe delivery to the designated point by common or other carrier at the lowest rate.

5.2.2 For export shipment.-

5.2.2.1 Place one each case of arrays, packaged as specified in 5.1.2.1, and one each bag of array accessories, packaged as specified in 5.1.2.2.4, together within a wood shipping container.

5.2.2.2 Pack two of each bags containing mast sections and stakes, packaged as specified in 5.1.2.3, and bags containing mast accessories, packaged as specified in 5.1.2.4.3, together within a wood shipping containers.

5.2.2.3 Wood shipping containers specified above shall be either wood cleated plywood boxes or nailed wood boxes conforming to Specifications JAN-P-105 or JAN-P-106 respectively. Fill all voids with cells and/or pads, fabricated of corrugated fiberboard, conforming to Specification 19-10, to prevent shifting. Fabricate to box to fit the contents snugly.

5.2.2.4 Close and strap the shipping container as specified in the appendix of the applicable box specification.

5.3 Marking for shipment.- Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD-129. On overseas shipping containers, the Signal Corps Insignia shall be placed on the lower left hand corner of the side containing contract data.

6. NOTES

6.1 The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example, Antenna AT-298W/GRC-10. As soon as possible after the award of the contract, the contractor should apply to the contract officer for such information.

6.2 Preparation for delivery.- Procurement documents should specify whether domestic or overseas preservation, packaging, and packing is required.

6.3 Contractual requirements.- It is recommended that the following be included in the contract.

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6.3.1 Additional testing.- It should be understood that additional inspection and non-destructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable documents.

6.3.2 Waiver of preproduction samples.- When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that (a) the specified samples have been submitted and approved on the current contract, (b) Production of the item under the new order will be concurrent with, or follow without interruption, production on the current order, and (c) current production is in strict compliance with specified requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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